

Jorge Becerra Garrido

Curriculum Vitae¹

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RESEARCH INTERESTS

I am mostly interested in quantum topology. The following is a non-exhaustive list of my interests:

- Quantum knot invariants, especially those which arise from ribbon categories/algebras.
- Tangle categories.
- Finite type invariants and the Kontsevich invariant.
- Modular categories, modular functors, modular operads.
- Factorisation homology, skein categories.

EMPLOYMENT

2026 – present	Profesor Ayudante Doctor , Universidad de La Rioja (Spain).
2024 – 2026	Postdoctoral fellow , Université Bourgogne Europe (France). Position in the research group of L. Woike.

EDUCATION

2019 – 2024	Ph.D. in Mathematics , <i>University of Groningen</i> (The Netherlands). Thesis: <i>Universal quantum knots invariants</i> , supervised by R. van der Veen.
2017 – 2019	M.Sc. Mathematical Sciences , <i>Utrecht University</i> (The Netherlands). Honours programme UGC, Cum Laude. Thesis: <i>K-theory with Reality</i> , supervised by L. Meier.
2013 – 2017	B.Sc. Mathematics , <i>University of Extremadura</i> (Spain). Thesis: <i>Theory of Relativity</i> , supervised by J.A. Navarro.

PUBLICATIONS & PREPRINTS

1. *Knot invariants from XC-structures on the Sweedler algebra are trivial*. Preprint.
2. *Pivotal Module Categories, Factorization Homology and Modular Invariant Modified Traces* (with Lukas Woike). Accepted for publication in Math. Z.
3. *XC-tangles and universal invariants*. Preprint.
4. *Minimal generating sets of rotational Reidemeister moves* (with Kevin van Helden). Preprint.
5. *A refined functorial universal tangle invariant*. Preprint.
6. *Strictification and non-strictification of monoidal categories*. Preprint.
7. *On Bar-Natan - van der Veen's perturbed Gaussians*, Rev. Real Acad. Cienc. Exactas Fis. Nat. Ser. A-Mat., 118(2):46, 2024.
8. *A combinatorial PROP for bialgebras*. European J. Combin. 124 (2025), Paper No. 104086.

¹Last updated: 21 September 2026

9. *An Introduction to Knot Homology Theories*, TEMat monográficos, 2 (2021): Proceedings of the 3rd BYMAT Conference, pp. 11-14. ISSN: 2660-6003.
10. *Finite, Infinite and Twisted Ascent and Descent* (with Jesús MF Castillo), La Gaceta de la Real Sociedad Matemática Española, Vol. 21 (2018), Núm. 3, Págs 497–516.

CONFERENCE TALKS

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| Jun 2026 | <i>Catégories module, homologie de factorisation et algèbres de Frobenius</i> [Module categories, factorisation homology and Frobenius algebras], Journée Scientifique de l'IMB , Dijon. |
| Jun 2026 | <i>Una fuente topológica de categorías módulo pivotaes</i> [A topological source of pivotal module categories], Métodos Categóricos y Homotópicos en Álgebra, Geometría, Topología, y Análisis Funcional 2026 , Murcia. |
| Jun 2026 | <i>Homotopy groups of spheres, framed cobordism, part I</i> , Matemale 2026 . |
| May 2026 | <i>The Witten-Reshetikhin-Turaev 3-manifold invariant</i> , Topology Between the Waves VI , Fréjus. |
| Jan 2026 | <i>XC-algebras and quantum knot invariants</i> , Congreso Bienal de la Real Sociedad Matemática Española , Alicante. |
| May 2025 | <i>Lee homology and the Lee spectral sequence</i> , Matemale 2025 . |
| May 2024 | <i>Universal quantum knot invariants</i> , Groningen Quantum Topology Dag . |
| Nov 2022 | <i>Three faces of the 2-loop polynomial of knots</i> , 4th BYMAT , online. |
| Aug 2022 | <i>Universal quantum knot invariants</i> , GQT School & Colloquium , Zeist. |
| Dec 2021 | <i>Quantum invariants via perturbed Gaussian expressions</i> , Winter Braids XI , Dijon. |
| Nov 2021 | <i>A categorical approach to the universal tangle invariant</i> , DIAMANT symposium , Utrecht. |
| Dec 2020 | <i>An introduction to knot homology theories</i> , 3rd BYMAT , online. |
| Mar 2017 | <i>Functional inequalities and convergence in diffusion processes</i> , 13th Meeting on Functional Analysis and Applications , Cáceres. |

SEMINAR TALKS

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| May 2026 | <i>XC-algebras and quantum knot invariants</i> , Topology Seminar , Genève. |
| May 2026 | <i>Construction of modular tensor categories from Hopf algebras III</i> , GDT sur TQFTs (non-)semisimples . |
| Apr 2026 | <i>Construction of modular tensor categories from Hopf algebras II</i> , GDT sur TQFTs (non-)semisimples . |
| Apr 2026 | <i>Construction of modular tensor categories from Hopf algebras I</i> , GDT sur TQFTs (non-)semisimples . |
| Feb 2026 | <i>Invariants des nœuds à partir des algèbres XC</i> [Knot invariants from XC-algebras], Séminaire de doctorants UBE . |
| Nov 2025 | <i>Classification des TQFT en dimension 1 et 2, II</i> [Classification of 1d and 2d TQFTs], GDT sur TQFTs (non-)semisimples . |
| Nov 2025 | <i>Classification des TQFT en dimension 1 et 2, I</i> [Classification of 1d and 2d TQFTs], GDT sur TQFTs (non-)semisimples . |
| Oct 2025 | <i>Lee homology and the Lee spectral sequence</i> , UIC Quantum Topology Seminar , Chicago (online). |
| Oct 2025 | <i>Le théorème d'Eilenberg-Watts et catégories finies</i> [The Eilenberg-Watts theorem and finite categories], Séminaire de doctorants UBE . |

Jan 2025	<i>How to cook (up) your own knot invariant from your favourite group</i> , Séminaire de doctorants UBE (based on pages 78–89 here).
Nov 2024	<i>Hopf-algebraic quantum knot invariants</i> , GADT groupe de travail , Dijon.
Jan 2024	<i>Hopf algebras and knot invariants</i> , ICMAT Geometry Seminar , Madrid.
Jan 2024	<i>Drinfeld doubles and Gaussian calculus for universal knot invariants</i> , MAPHYAG Seminar , Madrid.
Jan 2024	<i>Quantum invariants of knots and tangles</i> , IMUEx Seminar , Badajoz.
May 2023	<i>The Kontsevich invariant II</i> , Groningen.
May 2023	<i>The Kontsevich invariant I</i> , Groningen.
Dec 2022	<i>Hopf algebras and the 2-loop polynomial of knots</i> , Algebra Seminar , Groningen.
Nov 2022	<i>Classification of Riemann surfaces</i> , Riemann surfaces seminar, Groningen.
Oct 2022	<i>Categorification of finite type knot invariants</i> , GT Seminar , Groningen.
Jan 2022	<i>Cobordism and Thom spectra</i> , GT Seminar , Groningen.
Nov 2021	<i>Slice knots and knot concordance</i> , GT Seminar , Groningen.
Oct 2021	<i>My name is Theory, Category Theory</i> , Basic Notions seminar , Groningen.
May 2021	<i>Principal bundles from a topologist's point of view</i> , PhD Seminar , Groningen.
Nov 2020	<i>Khovanov homology à la Bar-Natan</i> , KHT Seminar , Groningen.
Oct 2020	<i>Isotopy invariance of Khovanov homology II</i> , KHT Seminar , Groningen.
Oct 2020	<i>Isotopy invariance of Khovanov homology I</i> , KHT Seminar , Groningen.
Sep 2020	<i>Overview of knot Floer and Khovanov homology</i> , KHT Seminar , Groningen.
May 2019	<i>Foundations of stable homotopy theory</i> , IMUEx Seminar , Badajoz.
Apr 2019	<i>Examples of Sullivan models</i> , Master Seminar on Rational Homotopy Groups , Utrecht.
Feb 2019	<i>Simplicial Sets</i> , Master Seminar on Rational Homotopy Groups , Utrecht.
May 2018	<i>Fredholm Operators and K-Theory</i> , Master Seminar on K-Theory , Utrecht.
Apr 2018	<i>The Hopf Invariant One Problem</i> , Master Seminar on K-Theory , Utrecht.
Mar 2018	<i>The Product Theorem</i> , Master Seminar on K-Theory , Utrecht.

RESEARCH VISITS

Jan 2024	Universidad Complutense de Madrid, invited by Ángel González Prieto (1 week).
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SELECTED ATTENDANCE

Jun 2026	<i>Métodos Categóricos y Homotópicos en Álgebra, Geometría, Topología, y Análisis Funcional 2026</i> , Murcia.
Jun 2026	<i>Matemale spring school 2026</i> , Pyrénées-Orientales.
May 2026	<i>Topology Between the Waves VI</i> , Fréjus.
Mar 2026	<i>Journées de topologie algébrique, géométrique et quantique en Picardie III</i> , Amiens.
Mar 2026	<i>Winter Braids XV</i> , Marseille.
Jan 2026	<i>Congreso Bienal de la RSME</i> , Alicante.
Sep 2025	<i>Higher Structures: Recent developments and applications</i> , Hamburg.
May 2025	<i>Matemale spring school 2025</i> , Matemale.
Mar 2025	<i>Algebraic, geometric and quantum topology in Picardie II</i> , Amiens.
Feb 2025	<i>Winter Braids XIV</i> , Bordeaux.

Dec 2024	<i>Skein days</i> , Cergy-Pontoise.
Mar 2024	<i>Algebraic, geometric and quantum topology in Picardie</i> , Amiens.
Sep 2023	<i>Swiss Knots 2023</i> , Regensburg.
Jun 2022	<i>From Subfactors to Quantum Topology - In memory of Vaughan Jones</i> , Geneva.
Jun 2022	<i>Quantum Topology and Geometry (in honor of Vladimir Turaev)</i> , Paris.
Dec 2021	<i>Winter Braids XI</i> , Dijon.
Jul 2021	<i>Georgia Tech Topology Summer School on 4-manifolds</i> , Online.
Dec 2020	<i>Monoidal and 2-categories in representation theory and categorification</i> , Online.
Aug 2020	<i>QUAntum groups, Categorification, Knot invariants, and Soergel bimodules</i> , Online.

TEACHING

Universidad de La Rioja

Autumn 2026	Lecturer of <i>Matemáticas I</i> (Bachelor).
Autumn 2026	Lecturer of <i>Geometría y Topología de Superficies</i> (Bachelor).

Université Bourgogne Europe

Spring 2026	Teaching Assistant for <i>Mathématiques pour l'électr. et l'inform. 2</i> (Bachelor).
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University of Groningen

Spring 2023	Co-lecturer of <i>Topics in Topology</i> (Master, universal knot invariants).
Spring 2022	Co-lecturer of <i>Topics in Topology</i> (Master, Kirby calculus).
Spring 2021	Co-lecturer of <i>Topics in Topology</i> (Master, grid homology).
Autumn 2020	Teaching Assistant for <i>Multivariable Analysis</i> (Bachelor).
Spring 2020	Co-organiser of the <i>Knot Theory Seminar</i> (Bach./Master).
Autumn 2019	Co-organiser of the <i>Groningen Topology Seminar</i> (Bach./Master).

Utrecht University

Spring 2019	Teaching Assistant for <i>Topologie en Meetkunde</i> (Bachelor, π_1).
Autumn 2018	Teaching Assistant for <i>Algebraic Topology</i> (Master, H_n 's and π_n 's).
Spring 2018	Teaching Assistant for <i>Topologie en Meetkunde</i> (Bachelor, π_1).

BACHELOR'S THESES SUPERVISION

2021	Daniël Apol, <i>Representability of cohomology theories on CW-complexes</i> , Groningen.
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SKILLS

- **Languages:** Spanish (native), English (fluent), French (intermediate), Dutch (basic).
- **Programming languages:**
Python (with [verified certificate](#) by HarvardX), including SAGE, Numpy, Pandas, Matplotlib, NetworkX and Scikit-learn.
Mathematica, C, R-Commander, Fortran, L^AT_EX, Linux.
- **Teaching:** Teacher training certificate.

ORGANISATION

2025 – 2026	Co-organiser of the talk series <i>Outils algébriques en théorie conforme des champs</i> , Dijon.
2025 – 2026	Co-organiser of the <i>Groupe de Travail sur TQFTs (non-)semisimples</i> , Dijon.
May 2024	Co-organiser of the workshop <i>Groningen Quantum Topology Dag</i> .

2022	Co-organiser of the Quantum Mechanics bookclub , Groningen.
2020 – 2021	Co-organiser of the PhD Seminar , Groningen.
2020	Co-organiser of the Knot homology theories reading group , Groningen.

GRANTS

2016 – 2017	Research Assistant , Department of Mathematics, University of Extremadura (Badajoz, Spain). €2,000.
2016	Fee waiver for talented students, Summer School on Geometry , Utrecht (The Netherlands). €400.
2013 – 2016	Twelve university course tuition fee waivers for top student of a course. €1,200.
2013	One-year university tuition fee waiver for top high-school student. €1,000.

PROFESSIONAL ACTIVITIES

2014 – 2017	Speaker at various outreach activities, e.g. the European Researchers' Night at the University of Extremadura.
2013 – 2017	Private tutor of English, Mathematics and Physics for high school students.